

Competitive Programming

Section 37 Editorial: Segment Trees

Competitive Programming Course

2026-07-06

Table of contents

1	Editorial overview	3
2	A. Range Minimum Updates	4
2.1	Restatement	4
2.2	Algorithm	4
2.3	Correctness proof	4
2.4	Complexity	4
2.5	Full C++ solution	4
2.6	Full Python solution	5
3	B. Maximum Subarray Updates	7
3.1	Restatement	7
3.2	Algorithm	7
3.3	Correctness proof	7
3.4	Complexity	7
3.5	Full C++ solution	7
3.6	Full Python solution	8
4	C. Lazy Range Add Sum	10
4.1	Restatement	10
4.2	Algorithm	10
4.3	Correctness proof	10
4.4	Complexity	10
4.5	Full C++ solution	10
4.6	Full Python solution	12
5	Testing discussion	14

1 Editorial overview

Problem	Main idea
A. Range Minimum Updates	See full algorithm and code below.
B. Maximum Subarray Updates	See full algorithm and code below.
C. Lazy Range Add Sum	See full algorithm and code below.

2 A. Range Minimum Updates

2.1 Restatement

Point-assign values and answer range minimum queries.

2.2 Algorithm

Identify the operation that must be fast, store exactly the aggregate information needed for that operation, and update the affected logarithmic frontier. The lesson for this section explains the invariant in detail; the implementation below is a direct transcription of that invariant.

For this problem, every query is processed in the order given except offline queries explicitly marked as offline. The answer is emitted immediately when the operation asks for one.

2.3 Correctness proof

The maintained structure stores the exact summary promised by the lesson: prefix blocks for Fenwick trees, segment aggregates for segment trees, power-of-two jumps for static structures, or the active frontier for heap/Mo-style processing. Each update changes precisely the summaries whose represented set contains the changed element or interval. Each query decomposes the requested object into summaries that are disjoint, idempotently overlapping, or maintained as the current active window. The merge formula for those summaries is exactly the mathematical definition of the requested answer, so every printed value is correct. \square

2.4 Complexity

The optimized operations are logarithmic unless the lesson says the structure is static or offline. Memory is linear or $O(n \log n)$ for sparse-table/binary-lifting preprocessing.

2.5 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);
    int n, q;
```

```

if (!(cin >> n >> q)) return 0;
int size = 1;
while (size < n) size <= 1;
const long long INF = (1LL << 60);
vector<long long> seg(2 * size, INF);
for (int i = 0; i < n; i++) cin >> seg[size + i];
for (int i = size - 1; i >= 1; i--) seg[i] = min(seg[2 * i], seg[2 * i + 1]);
while (q--) {
    int typ, x, y;
    cin >> typ >> x >> y;
    if (typ == 1) {
        int p = size + x - 1;
        seg[p] = y;
        for (p >= 1; p; p >= 1) seg[p] = min(seg[2 * p], seg[2 * p + 1]);
    } else {
        int l = size + x - 1, r = size + y;
        long long ans = INF;
        while (l < r) {
            if (l & 1) ans = min(ans, seg[l++]);
            if (r & 1) ans = min(ans, seg[--r]);
            l >>= 1;
            r >>= 1;
        }
        cout << ans << '\n';
    }
}
return 0;
}

```

2.6 Full Python solution

```

import sys

INF = 10**30

def main():
    data = list(map(int, sys.stdin.buffer.read().split()))
    if not data:
        return
    n, q = data[0], data[1]
    a = data[2:2 + n]
    size = 1
    while size < n:
        size *= 2
    seg = [INF] * (2 * size)
    for i, x in enumerate(a):
        seg[size + i] = x
    for i in range(size - 1, 0, -1):

```

```

        seg[i] = min(seg[2 * i], seg[2 * i + 1])
    idx = 2 + n
    out = []
    for _ in range(q):
        typ, x, y = data[idx], data[idx + 1], data[idx + 2]
        idx += 3
        if typ == 1:
            p = size + x - 1
            seg[p] = y
            p //= 2
            while p:
                seg[p] = min(seg[2 * p], seg[2 * p + 1])
                p //= 2
        else:
            l, r = x - 1 + size, y + size
            ans = INF
            while l < r:
                if l & 1:
                    ans = min(ans, seg[l])
                    l += 1
                if r & 1:
                    r -= 1
                    ans = min(ans, seg[r])
                l //= 2
                r //= 2
            out.append(str(ans))
    print("\n".join(out))

if __name__ == "__main__":
    main()

```

3 B. Maximum Subarray Updates

3.1 Restatement

After each point assignment, print the maximum non-empty subarray sum of the whole array.

3.2 Algorithm

Identify the operation that must be fast, store exactly the aggregate information needed for that operation, and update the affected logarithmic frontier. The lesson for this section explains the invariant in detail; the implementation below is a direct transcription of that invariant.

For this problem, every query is processed in the order given except offline queries explicitly marked as offline. The answer is emitted immediately when the operation asks for one.

3.3 Correctness proof

The maintained structure stores the exact summary promised by the lesson: prefix blocks for Fenwick trees, segment aggregates for segment trees, power-of-two jumps for static structures, or the active frontier for heap/Mo-style processing. Each update changes precisely the summaries whose represented set contains the changed element or interval. Each query decomposes the requested object into summaries that are disjoint, idempotently overlapping, or maintained as the current active window. The merge formula for those summaries is exactly the mathematical definition of the requested answer, so every printed value is correct. $\square$

3.4 Complexity

The optimized operations are logarithmic unless the lesson says the structure is static or offline. Memory is linear or $O(n \log n)$ for sparse-table/binary-lifting preprocessing.

3.5 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;

const long long NEG = -(1LL << 60);

struct Node {
    long long sum, pref, suff, best;
```

```

};

Node leaf(long long x) { return {x, x, x, x}; }
Node merge(Node a, Node b) {
    return {
        a.sum + b.sum,
        max(a.pref, a.sum + b.pref),
        max(b.suff, b.sum + a.suff),
        max({a.best, b.best, a.suff + b.pref})
    };
}

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);
    int n, q;
    if (!(cin >> n >> q)) return 0;
    int size = 1;
    while (size < n) size <= 1;
    Node neutral{0, NEG, NEG, NEG};
    vector<Node> seg(2 * size, neutral);
    for (int i = 0; i < n; i++) {
        long long x;
        cin >> x;
        seg[size + i] = leaf(x);
    }
    for (int i = size - 1; i >= 1; i--) seg[i] = merge(seg[2 * i], seg[2 * i + 1]);
    while (q--) {
        int pos;
        long long val;
        cin >> pos >> val;
        int p = size + pos - 1;
        seg[p] = leaf(val);
        for (p >= 1; p; p >= 1) seg[p] = merge(seg[2 * p], seg[2 * p + 1]);
        cout << seg[1].best << '\n';
    }
    return 0;
}

```

3.6 Full Python solution

```

import sys

NEG = -10**30

def merge(a, b):
    total = a[0] + b[0]
    pref = max(a[1], a[0] + b[1])

```

```

    suff = max(b[2], b[0] + a[2])
    best = max(a[3], b[3], a[2] + b[1])
    return (total, pref, suff, best)

def leaf(x):
    return (x, x, x, x)

def main():
    data = list(map(int, sys.stdin.buffer.read().split()))
    if not data:
        return
    n, q = data[0], data[1]
    a = data[2:2 + n]
    size = 1
    while size < n:
        size *= 2
    neutral = (0, NEG, NEG, NEG)
    seg = [neutral] * (2 * size)
    for i, x in enumerate(a):
        seg[size + i] = leaf(x)
    for i in range(size - 1, 0, -1):
        seg[i] = merge(seg[2 * i], seg[2 * i + 1])
    idx = 2 + n
    out = []
    for _ in range(q):
        pos, val = data[idx], data[idx + 1]
        idx += 2
        p = size + pos - 1
        seg[p] = leaf(val)
        p //= 2
        while p:
            seg[p] = merge(seg[2 * p], seg[2 * p + 1])
            p //= 2
        out.append(str(seg[1][3]))
    print("\n".join(out))

if __name__ == "__main__":
    main()

```

4 C. Lazy Range Add Sum

4.1 Restatement

Support range add updates and range sum queries.

4.2 Algorithm

Identify the operation that must be fast, store exactly the aggregate information needed for that operation, and update the affected logarithmic frontier. The lesson for this section explains the invariant in detail; the implementation below is a direct transcription of that invariant.

For this problem, every query is processed in the order given except offline queries explicitly marked as offline. The answer is emitted immediately when the operation asks for one.

4.3 Correctness proof

The maintained structure stores the exact summary promised by the lesson: prefix blocks for Fenwick trees, segment aggregates for segment trees, power-of-two jumps for static structures, or the active frontier for heap/Mo-style processing. Each update changes precisely the summaries whose represented set contains the changed element or interval. Each query decomposes the requested object into summaries that are disjoint, idempotently overlapping, or maintained as the current active window. The merge formula for those summaries is exactly the mathematical definition of the requested answer, so every printed value is correct. $\square$

4.4 Complexity

The optimized operations are logarithmic unless the lesson says the structure is static or offline. Memory is linear or $O(n \log n)$ for sparse-table/binary-lifting preprocessing.

4.5 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;

struct SegTree {
    int n;
    vector<long long> seg, lazy;
    SegTree(const vector<long long>& a) {
```

```

    n = 1;
    while (n < (int)a.size()) n <= 1;
    seg.assign(2 * n, 0);
    lazy.assign(2 * n, 0);
    for (int i = 0; i < (int)a.size(); i++) seg[n + i] = a[i];
    for (int i = n - 1; i >= 1; i--) seg[i] = seg[2 * i] + seg[2 * i + 1];
}

void apply(int v, int len, long long x) {
    seg[v] += x * len;
    lazy[v] += x;
}

void push(int v, int len) {
    if (lazy[v] && v < n) {
        int half = len / 2;
        apply(2 * v, half, lazy[v]);
        apply(2 * v + 1, half, lazy[v]);
        lazy[v] = 0;
    }
}

void add(int v, int tl, int tr, int l, int r, long long x) {
    if (l <= tl && tr <= r) {
        apply(v, tr - tl + 1, x);
        return;
    }
    push(v, tr - tl + 1);
    int tm = (tl + tr) / 2;
    if (l <= tm) add(2 * v, tl, tm, l, r, x);
    if (tm < r) add(2 * v + 1, tm + 1, tr, l, r, x);
    seg[v] = seg[2 * v] + seg[2 * v + 1];
}

long long query(int v, int tl, int tr, int l, int r) {
    if (l <= tl && tr <= r) return seg[v];
    push(v, tr - tl + 1);
    int tm = (tl + tr) / 2;
    long long ans = 0;
    if (l <= tm) ans += query(2 * v, tl, tm, l, r);
    if (tm < r) ans += query(2 * v + 1, tm + 1, tr, l, r);
    return ans;
}

};

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);
    int n, q;
    if (!(cin >> n >> q)) return 0;
    vector<long long> a(n);
    for (long long &x : a) cin >> x;
    SegTree st(a);
    while (q--) {
        int typ, l, r;

```

```

    cin >> typ >> l >> r;
    if (typ == 1) {
        long long x;
        cin >> x;
        st.add(1, 1, st.n, l, r, x);
    } else {
        cout << st.query(1, 1, st.n, l, r) << '\n';
    }
}
return 0;
}

```

4.6 Full Python solution

```

import sys

def main():
    data = list(map(int, sys.stdin.buffer.read().split()))
    if not data:
        return
    n, q = data[0], data[1]
    a = data[2:2 + n]
    size = 1
    while size < n:
        size *= 2
    seg = [0] * (2 * size)
    lazy = [0] * (2 * size)
    for i, x in enumerate(a):
        seg[size + i] = x
    for i in range(size - 1, 0, -1):
        seg[i] = seg[2 * i] + seg[2 * i + 1]
    def apply(v, length, delta):
        seg[v] += delta * length
        lazy[v] += delta
    def push(v, length):
        if lazy[v] and v < size:
            half = length // 2
            apply(2 * v, half, lazy[v])
            apply(2 * v + 1, half, lazy[v])
            lazy[v] = 0
    def add(v, tl, tr, l, r, x):
        if l <= tl and tr <= r:
            apply(v, tr - tl + 1, x)
            return
        push(v, tr - tl + 1)
        tm = (tl + tr) // 2
        if l <= tm:
            add(2 * v, tl, tm, l, r, x)
        if tm < r:

```

```

        add(2 * v + 1, tm + 1, tr, l, r, x)
    seg[v] = seg[2 * v] + seg[2 * v + 1]
def query(v, tl, tr, l, r):
    if l <= tl and tr <= r:
        return seg[v]
    push(v, tr - tl + 1)
    tm = (tl + tr) // 2
    ans = 0
    if l <= tm:
        ans += query(2 * v, tl, tm, l, r)
    if tm < r:
        ans += query(2 * v + 1, tm + 1, tr, l, r)
    return ans
idx = 2 + n
out = []
for _ in range(q):
    typ = data[idx]
    idx += 1
    if typ == 1:
        l, r, x = data[idx], data[idx + 1], data[idx + 2]
        idx += 3
        add(1, 1, size, l, r, x)
    else:
        l, r = data[idx], data[idx + 1]
        idx += 2
    out.append(str(query(1, 1, size, l, r)))
print("\n".join(out))

if __name__ == "__main__":
    main()

```

5 Testing discussion

The fixed samples cover visible behavior. The random tests generate many small operation sequences and compute expected outputs from the reference Python implementation, then the same expected files are used against both the C++ and Python submissions.